

UNIVERSITY OF SCIENCE - VNUHCM FACULTY
OF INFORMATION TECHNOLOGY



Software Design

Introduction to Software Engineering (CSC13002)

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1 Member Contribution Assessment

ID	Name	Contribution (%)	Signature
24127540	Lê Tấn Thành	100%	
24127234	Lê Song Anh Tài	100%	
23127008	Nguyễn Trọng Tài	98%	
24127479	Nguyễn Bách Nhuận	100%	

2 Conceptual Model



Figure. AI Content and Post Creation ERD

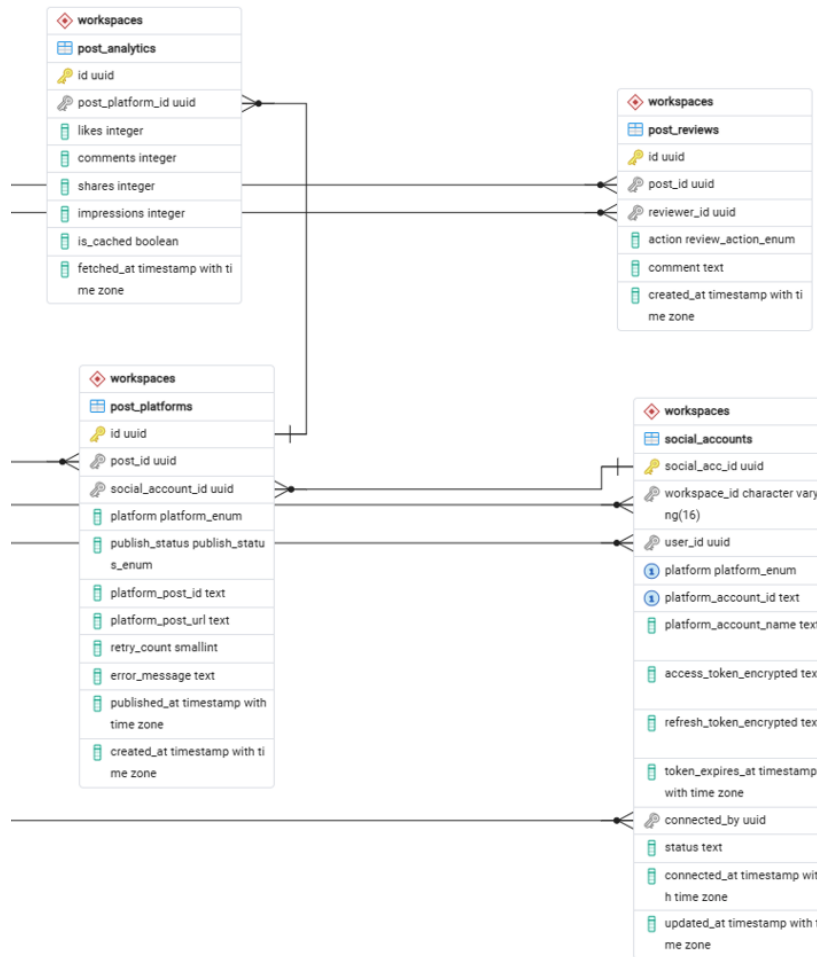


Figure. Review, Social Media Publishing, and Post Analytics ERD



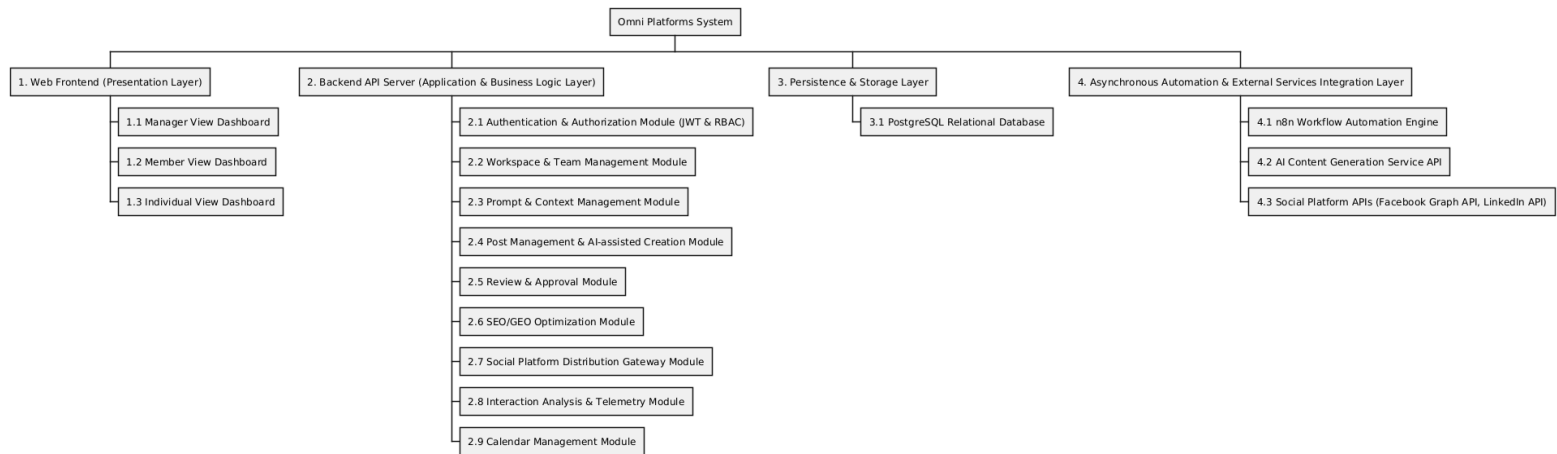
Figure. User and Workspace Membership ERD

[Click here for the High-Resolution Image](#)

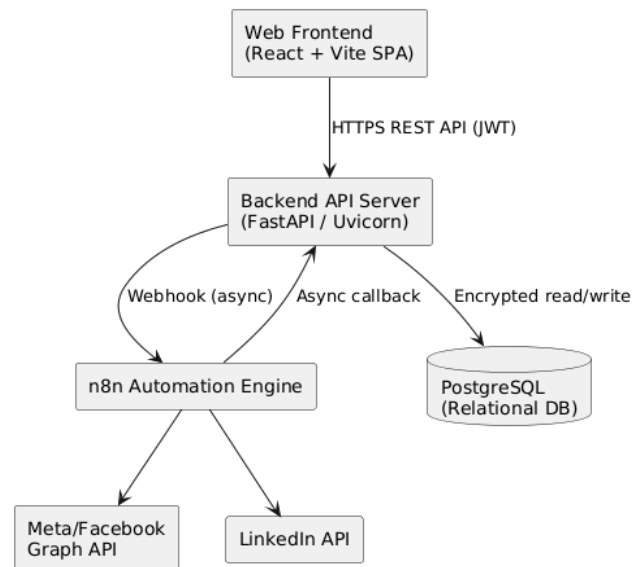
3 Architectural Design

3.1 Architecture Diagram

3.1.1 System Decomposition Tree Diagram



3.1.2 Overall System Architecture Diagram



3.1.3 Special Architectural Aspects & Design Patterns

1. Decoupled Client-Server & Layered Architecture

The system adopts a fully decoupled client-server architecture, where the presentation layer (Client) and the business logic layer (Server) communicate exclusively through a standardized RESTful API over HTTPS. The two layers are independent of each other in terms of implementation technologies.

In Omni Platforms, the Web Frontend (React + Vite) contains no business logic or authorization rules. All business processes, validation, and access control are enforced exclusively by the Backend (FastAPI). This separation allows the frontend to be replaced entirely without requiring any modifications to the backend codebase.

2. Asynchronous Task Delegation (n8n Engine)

The system employs an asynchronous task delegation pattern to offload high-latency, I/O-intensive operations from the main request-processing flow, preventing blocking and improving system responsiveness.

When a post is approved and ready for publication, the Backend does not directly invoke external social media APIs, whose response times may be unpredictable. Instead, it sends a webhook payload to the n8n Automation Engine. To handle potential execution errors without tight coupling, the architecture implements a Two-Tier Retry Strategy:

- **Transient Retry (n8n Layer):** n8n handles immediate network or rate-limiting glitches automatically using an exponential backoff policy during the webhook execution flow.
- **Persistent/Scheduled Retry (Backend Layer):** If n8n exhausts its retry attempts, an error callback updates the post status to FAILED and logs the failure details into PostPlatform. The RetryScheduler in the Backend then manages long-term retry queues (e.g., after user re-authentication or via scheduled cron jobs).

Once processing completes, n8n transmits the execution result back to the Backend to update PostPlatform. This architecture ensures that the Backend consistently

maintains a response time below 500 ms regardless of third-party platform performance.

3. Role-Based Access Control (RBAC) Pattern

The system implements the Role-Based Access Control (RBAC) pattern by centralizing authorization logic within a dedicated middleware layer instead of distributing permission checks across individual application functions.

In Omni Platforms, the FastAPI RBAC Middleware decodes the JWT token generated by the `generate_jwt_token()` method of the `User` class to determine whether the authenticated user has the Manager, Member, or Individual role before granting access to protected resources. Authorization constraints such as `submit_for_review()` (Member only) and `remove_member()` (Manager only), as specified in the Constraint column of the Operations table in Section 3.3, are enforced by this middleware. This approach follows the Security by Design principle by ensuring that authorization decisions are made exclusively on the server side and never rely on client-side validation.

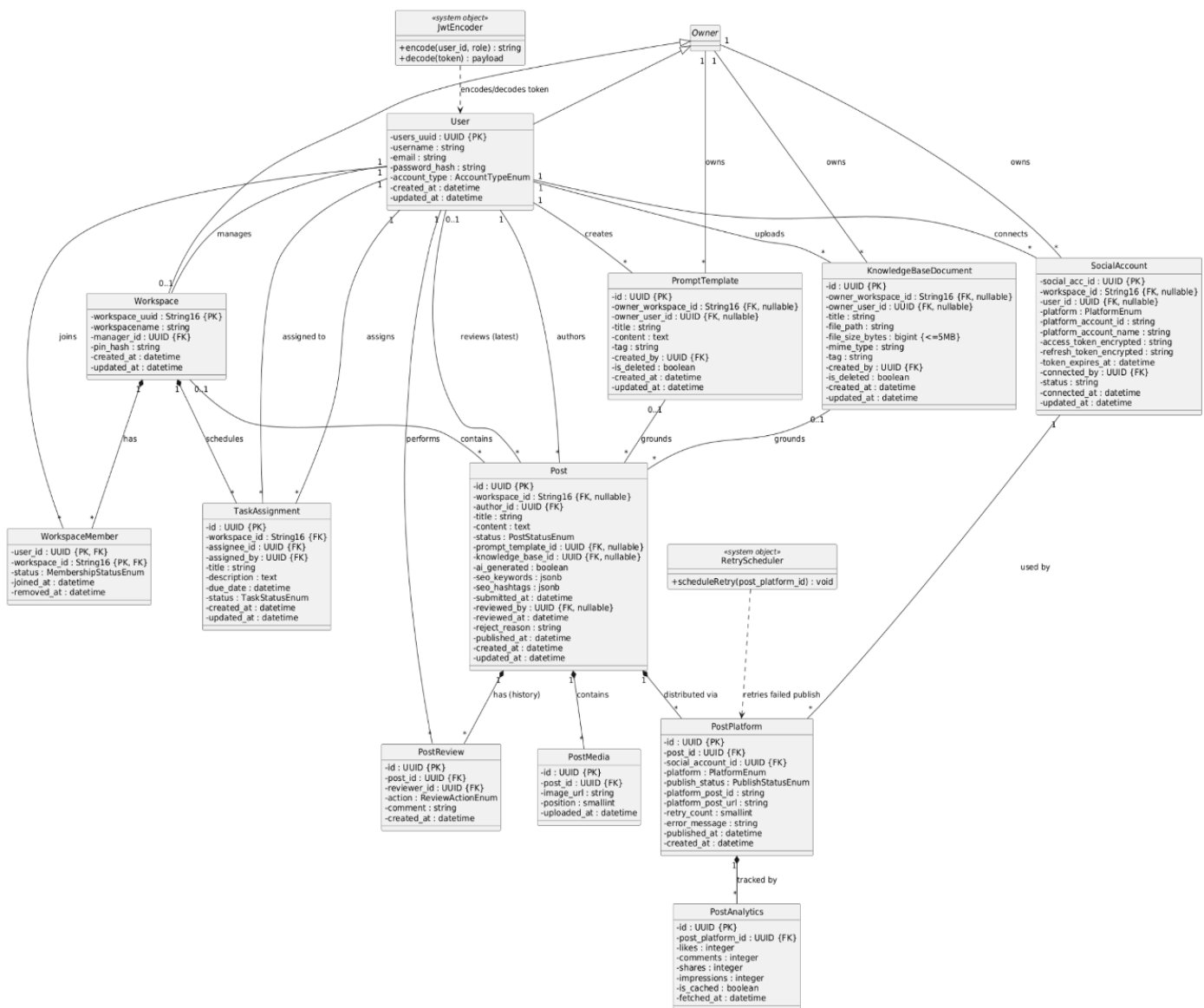
4. Repository/DAO Pattern

The system applies the Repository/DAO pattern to separate data access logic from business logic, allowing the business layer to remain independent of database-specific implementation details.

Entity classes such as `Post`, `User`, `ReviewLog`, and `OAuthToken` do not execute SQL statements directly. Instead, all Create, Read, Update, and Delete (CRUD) operations are performed through SQLAlchemy ORM, which serves as the repository layer. This abstraction improves maintainability and portability by enabling the underlying database management system (e.g., PostgreSQL) to be replaced with another

database technology in the future with minimal impact on the business logic implemented in FastAPI.

3.2 Class Diagram



3.3 Class Specifications

3.3.1 Class User

Seq	Property	Modifier	Constraint	Description
1	users_uuid	private	PK	Unique identifier
2	username	private		Username
3	email	private	Unique	Contact email
4	password_hash	private		Hashed password
5	account_type	private	Enum (individual/business)	Account type
6	created_at	private		Creation timestamp
7	updated_at	private		Update timestamp

Seq	Operation	Modifier	Constraint	Description
1	generate_jwt_token()	public		Generate JWT containing role inferred from workspace

2	verify_password()	public		Verify password upon login
3	update_profile()	public	Owner only	Update profile information

3.3.2 Class Workspace

Seq	Property	Modifier	Constraint	Description
1	workspace_uuid	private	PK	Workspace identifier
2	workspacename	private		Workspace name
3	manager_id	private	FK → User, UNIQUE	Manager (each user can only manage one workspace)
4	pin_hash	private		Hashed PIN
5	created_at	private		Creation timestamp
6	updated_at	private		Update timestamp

Seq	Operation	Modifier	Constraint	Description
1	add_member()	public	Manager only	Add member
2	remove_member()	public	Manager only	Remove member

3.3.3 Class WorkspaceMember

Seq	Property	Modifier	Constraint	Description
-----	----------	----------	------------	-------------

1	user_id	private	PK, FK → User	User identifier
2	workspace_id	private	PK, FK → Workspace	Workspace identifier
3	status	private	Enum (pending/active/removed)	Member status
4	joined_at	private		Join timestamp
5	removed_at	private	Nullable	Removal timestamp

Seq	Operation	Modifier	Constraint	Description
1	update_status()	public	Manager only	Update member status

3.3.4 Class TaskAssignment

Seq	Property	Modifier	Constraint	Description
1	id	private	PK	Unique identifier for the task
2	workspace_id	private	FK → Workspace	Foreign key linking to the corresponding workspace

3	assignee_id	private	FK → User	The user assigned to perform the task
4	assigned_by	private	FK → User	The user who created or assigned the task
5	title	private		Title or name of the task
6	description	private		Detailed description of the task content
7	due_date	private	Nullable	Deadline for task completion
8	status	private	Enum (pending/in_progress/done)	Current state or progress of the task

Seq	Operation	Modifier	Constraint	Description
1	assign_task()	public	Manager only	Assign a task to a team member
2	update_status()	public	Assignee only	Update the progress/status of the assigned task

3.3.5 Class Post

Seq	Property	Modifier	Constraint	Description
-----	----------	----------	------------	-------------

1	id	private	PK	Unique identifier for the post
2	workspace_id	private	FK, Nullable	Foreign key linking to the workspace (if applicable)
3	author_id	private	FK → User	The user who created the post
4	title,	private		Title of the post
5	Content			Main content of the post
6	status	private	Enum (draft/pending/approved/rejected/published)	Current status of the post workflow
7	prompt_template_id	private	FK, Nullable	Foreign key linking to the AI prompt template used
8	knowledge_base_id	private	FK, Nullable	Foreign key linking to the reference knowledge base
9	ai_generated	private		Boolean flag indicating if the post was generated by AI
9	seo_keywords	private	jsonb	Keywords stored in JSON format for SEO optimization

10	seo_hashtags	private	jsonb	Hashtags stored in JSON format for SEO optimization
11	Reviewed_by	private	FK, Nullable	Foreign key linking to the user who reviewed the post
12	reject_reason	private	Nullable	Reason for rejecting the post (if rejected)

Seq	Operation	Modifier	Constraint	Description
1	submit_for_review()	public	Member only	Submit the post for approval
2	approve()	public	Manager only	Approve the submitted post
3	reject()	public	Manager only	Reject the post and record the reason
4	publish()	public		Trigger/Activate post distribution

3.3.6 Class PostReview

Seq	Property	Modifier	Constraint	Description
1	id	private	PK	Unique identifier for the review record

2	post_id	private	FK → Post	Foreign key linking to the reviewed post
3	reviewer_id	private	FK → User	Foreign key linking to the user who performed the review
4	action	private	Enum (approve/reject)	Action taken by the reviewer
5	comment	private	Nullable	Review feedback or reason for rejection
6	created_at	private		Timestamp when the review was created

Seq	Operation	Modifier	Constraint	Description
1	record_review()	public	Reviewer only	log the review decision and comments

3.3.7 Class PromptTemplate

Seq	Property	Modifier	Constraint	Description
Seq	Property	Modifier	Constraint	Description
1	id	private	PK	Unique identifier for the prompt template

2	owner_workspace_id / owner_user_id	private	FK, Nullable, mutually exclusive	Foreign key linking to the owner workspace
3	title	private		Name or title of the prompt template
4	content	private		The prompt content/instructions template
5	tag	private		Category or label for filtering prompt templates
6	created_by	private	FK → User	The user who created the template
7	is_deleted	private	Soft delete	Flag indicating whether the template is soft-deleted

Seq	Operation	Modifier	Constraint	Description
1	create_template()	public		Create a new prompt template
2	delete()	public	Owner only	Perform a soft delete on the template

3.3.8 Class KnowledgeBaseDocument

Seq	Property	Modifier	Constraint	Description
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1	id	private	PK	Unique identifier for the knowledge base document (Primary Key).
2	owner_workspace_id / owner_user_id	private	FK, Nullable, mutually exclusive	Foreign key referencing the owning Workspace ID or User ID. Exactly one must be set to determine whether the document is a shared workspace asset or a personal document.
3	file_path, file_size_bytes, mime_type	private	≤5MB	Storage file path, file size in bytes (maximum 5MB), and the MIME media type
4	created_by	private	FK → User	Foreign key referencing the User who uploaded the document to the system.

Seq	Operation	Modifier	Constraint	Description
1	upload()	public		Uploads a document file to storage

2	delete()	public	Owner only	Soft delete
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3.3.9 Class SocialAccount

Seq	Property	Modifier	Constraint	Description
1	social_acc_id	private	PK	Unique identifier for the social account record
2	workspace_id / user_id	private	FK, Nullable, mutually exclusive	Foreign key referencing either the Workspace ID or User ID. Exactly one must be set to indicate whether this connected social account belongs to a workspace or an individual user.
3	platform	private	Enum	The social media platform provider (e.g., Facebook, LinkedIn, Twitter/X, Instagram).
4	access_token_encrypted, refresh_token_encrypted	private	Encrypted	Securely encrypted OAuth tokens (access token and refresh token) used to authenticate API requests with the third-party platform.

5	token_expires_at	private		Timestamp indicating when the current OAuth access token will expire and require refreshing.
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Seq	Operation	Modifier	Constraint	Description
1	connect()	public		Connect to a social media account via OAuth authentication.
2	refresh_token()	public		Refreshes the OAuth access token using the stored refresh token when it expires.
3	disconnect()	public	Owner only	Revokes access tokens and disconnects the social account. Restricted to the owner of the user/workspace account.

3.3.10 Class PostPlatform

Seq	Property	Modifier	Constraint	Description
1	id	private	PK	Unique identifier for the post-to-platform

2	post_id, social_account_id	private	FK	Foreign keys referencing the target Post ID and the specific Social Account ID through which the post is published.
3	publish_status	private	Enum	The current status of the publishing task (e.g., Pending, Success, Failed).
4	retry_count, error_message	private		Tracks the number of publishing retry attempts and stores the failure error details if the API call fails.

Seq	Operation	Modifier	Constraint	Description
1	publish_to_platform()	public		Executes an API call to the respective social media platform to publish the post content.
2	retry_publish()	public		Retries publishing the post to the platform upon failure

3.3.11 Class PostAnalytics

Seq	Property	Modifier	Constraint	Description
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1	id	private	PK	Unique identifier for the analytics snapshot
2	post_platform_id	private	FK -> PostPlatform	Foreign key referencing the published platform record
3	likes	private	integer	Number of likes at the time of fetch
4	comments	private	integer	Number of comments at the time of fetch
5	shares	private	integer	Number of shares at the time of fetch
6	impressions	private	integer	Total view/impression count at the time of fetch
7	is_cached	private	boolean	Flag indicating whether metric is cached or fresh
8	fetches_at	private		Timestamp when performance metrics were fetched

Seq	Operation	Modifier	Constraint	Description
1	fetch_metrics()	public		Calls third-party API via

				n8n/gateway to pull real-time engagement data
2	get_cached_analytics()	public		Retrieves previously cached metrics for reporting and dashboard widgets

3.3.12 Class PostMedia

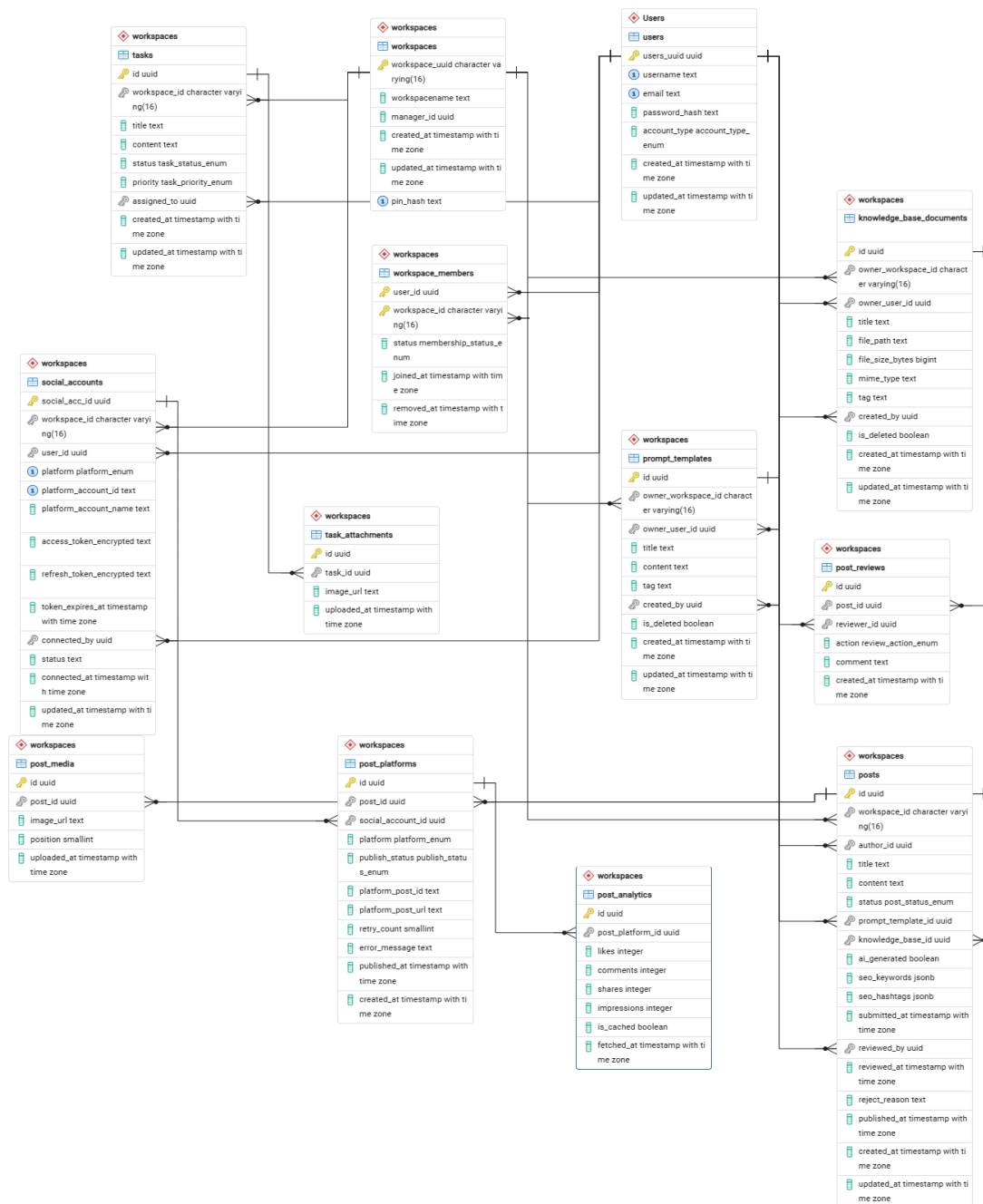
Seq	Property	Modifier	Constraint	Description
1	id	private	PK	Unique identifier for the media item
2	post_id	private	FK -> Post	Foreign key referencing the associated Post
3	image_url	private		Storage URL or path of the uploaded image/media
4	position	private	smallint	Display order of the media within the post gallery (0 = first)
5	uploaded_at	private	timestamp	Timestamp when the media was uploaded

Seq	Operation	Modifier	Constraint	Description
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1	upload_media()	public		Uploads media file to storage and links it to the specified post
2	delete_media()	public	Owner only	Removes the media item from storage and detaches it from the post

4 Data Design

4.1 Data Diagram



4.1.1 Data Diagram

4.2 Data Specification

4.2.1 Enum types

Enum Type	Component	Purpose
account_type_enum	individual, business	Distinguishes whether a user account operates as a personal (individual) account or a business account that owns workspaces.
membership_status_enum	pending, active, removed	Tracks the state of a user's membership inside a business workspace.
platform_enum	facebook, linkedin	Identifies which external social media platform a connected account or published post belongs to.
post_status_enum	draft, pending_review, rejected, ready_for_distribution, published, failed	Represents the workflow stage of a post, from creation to publication
publish_status_enum	pending, published, failed	Represents the outcome of attempting to push a single post to a single connected platform account.
review_action_enum	approve, reject	Records the decision a reviewer made when evaluating a submitted post.

4.2.2 Functions and Triggers

- `set_updated_at()`: a trigger function that sets `NEW.updated_at = now()` before any `UPDATE`.
- `generate_workspace_id()`: generates a random 16-character code from a 32-character alphabet that excludes visually ambiguous characters (l, L, O, 0, 1), then checks it against existing `workspace_uuid` values in a loop until a unique code is found.

4.2.3 Schema "Users"

Attribute	Data Type	Key Constraint	Nullable / Default	Description
users_uuid	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique, time-ordered identifier for the user; used as the FK target from all other tables that reference a user.
username	text	UNIQUE (case-insensitive)	NOT NULL	Login name. A unique functional index on <code>lower(username)</code> enforces case-insensitive uniqueness (e.g. "Alice" and "alice" cannot both exist).

email	text	UNIQUE (case-insensitive)	NOT NULL	Login email address. A unique functional index on lower(email) enforces case-insensitive uniqueness.
password_hash	text	—	NOT NULL	Argon2id hash of the user's password.
account_type	account_type_enum	—	NOT NULL, default 'individual'	Whether the account is a personal (individual) account or a business account; controls which registration/permission path the account follows.
created_at	timestamp	—	NOT NULL, default now()	Timestamp the account was created.
updated_at	timestamp	—	NOT NULL, default now()	Timestamp of the last update; auto-maintained by trg_users_updated_at.

4.2.4 Schema "workspaces"

4.2.4.1 workspaces

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
workspace_uuid	varchar (16)	PRIMARY_KEY, UNIQUE	NOT NULL, default generate_workspace_id()	16-character shareable workspace code; length is enforced by the workspace_uuid_length CHECK constraint.
workspace_name	text	—	NOT NULL	Display name of the business workspace.
manager_id	uuid	UNIQUE, FK → User	NOT NULL	Identifier of the user who manages the workspace (Each manager can only own 1 workspace)
pin_hash	text	UNIQUE	NOT NULL, default ''	Hash of an access PIN used to join into the workspace; each

				hash must be unique across all workspaces.
created_at	timestamp	—	NOT NULL, default now()	Timestamp the workspace was created.
updated_at	timestamp	—	NOT NULL, default now()	Timestamp of the last update; auto-maintained by trg_workspaces_updated_at.

4.2.4.2 workspace_members

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
user_id	uuid	- PRIMARY KEY user_id, workspace_id) - FK→"Users".users(users_uuid) - ON DELETE CASCADE	NOT NULL	The member's user account. Deleting a user removes their membership rows.
workspace_id	uuid	- PRIMARY KEY (composite) - FK→workspaces.workspaces (workspace_uuid) - ON DELETE CASCADE	NOT NULL	The workspace being joined. Deleting a workspace removes all of its membership rows.
status	membership_status_enum	—	NOT NULL, default 'active'	Current membership state: pending (invited/awaiting approval), active, or removed.
joined_at	timestamp	—	NOT NULL, default now()	Timestamp the membership record was created.
removed_at	timestamp	—	nullable	Timestamp the member was removed, if applicable; NULL while active or pending.

4.2.4.3 social_accounts

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
social_acc_id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique identifier for the connected social account record.
workspace_id	varchar(16)	- FK→workspaces.workspaces (workspace_uuid) - ON DELETE CASCADE	nullable	Owning workspace, if this is a workspace-level connection. Exactly one of workspace_id / user_id must be set (owner_exclusive CHECK constraint).
user_id	uuid	- FK→ "Users".users (users_uuid) - ON DELETE CASCADE	nullable	Owning individual user, if this is a personal connection. Exactly one of workspace_id / user_id must be set.
platform	platform_enum	—	NOT NULL	Which social platform this account belongs to (facebook or linkedin).
platform_account_id	text	UNIQUE together with platform	NOT NULL	The account's ID on the external platform; combined with platform, must be unique so the same external account cannot be connected twice.
platform_account_name	text	—	NOT NULL	Display name of the account as shown on the external platform.
access_token_encrypted	text	—	NOT NULL	Encrypted OAuth access token used to call the platform's API.

refresh_token_encrypted	text	—	nullable	Encrypted OAuth refresh token, if the platform issues one.
token_expires_at	timestamp	—	nullable	Expiry timestamp of the current access token.
connected_by	uuid	FK → "Users".users(users_uuid)	NOT NULL	The user who performed the OAuth connection.
status	text	—	NOT NULL, default 'connected'	Free-text connection status
connected_at	timestamp	—	NOT NULL, default now()	Timestamp the account was first connected.
updated_at	timestamp	—	NOT NULL, default now()	Timestamp of the last update to this record.

4.2.4.4 prompt_templates

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique identifier for the prompt template.
owner_workspace_id	varchar(16)	- FK→workspaces.workspaces (workspace_uuid) - ON DELETE CASCADE	nullable	Owning workspace, if shared at the workspace level. Exactly one of owner_workspace_id / owner_user_id must be set (owner_exclusive CHECK).

owner_user_id	uuid	- FK→ "Users".users (users_uuid) - ON DELETE CASCADE	nullable	Owning individual user, if personal. Exactly one of owner_workspace_id / owner_user_id must be set.
title	text	—	NOT NULL	Short label identifying the template.
content	text	—	NOT NULL	The actual prompt text sent to the AI generation service.
tag	text	—	nullable	Optional free-text tag for organizing templates.
created_by	uuid	FK → "Users".users(users_uuid)	NOT NULL	User who authored the template.
is_deleted	boolean	—	NOT NULL, default false	Soft-delete flag; a true value hides the template from normal listings without removing the row.
created_at	timestamp t	—	NOT NULL, default now()	Creation timestamp.
updated_at	timestamp t	—	NOT NULL, default now()	Last-update timestamp; auto-maintained by trg_prompt_templates_updated_at.

4.2.4.5 knowledge_base_documents

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique identifier for the document record.

owner_workspace_id	varchar(16)	- FK→workspaces.workspaces (workspace_uuid) - ON DELETE CASCADE	nullable	Owning workspace, if shared at the workspace level. Exactly one of owner_workspace_id / owner_user_id must be set (owner_exclusive CHECK).
owner_user_id	uuid	- FK → "Users".users(users_uuid) - ON DELETE CASCADE	nullable	Owning individual user, if personal. Exactly one of owner_workspace_id / owner_user_id must be set.
title	text	—	NOT NULL	Human-readable document title.
file_path	text	—	NOT NULL	Storage location/URL of the uploaded file.
file_size_bytes	bigint	CHECK file_size_limit	NOT NULL	File size in bytes; constrained to be at most 5 MiB
mime_type	text	—	NOT NULL	MIME type of the uploaded file
tag	text	—	nullable	Optional free-text tag.
created_by	uuid	FK → "Users".users(users_uuid)	NOT NULL	User who uploaded the document.
is_deleted	boolean	—	NOT NULL, default false	Soft-delete flag.
created_at	timestamp	—	NOT NULL, default now()	Creation timestamp.
updated_at	timestamp	—	NOT NULL, default now()	Last-update timestamp; auto-maintained by trg_kb_documents_updated_at.

4.2.4.6 posts

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique identifier for the post.
workspace_id	varchar(16)	- FK→workspaces.workspaces (workspace_uuid) - ON DELETE CASCADE	nullable	Workspace the post belongs to; NULL for a personal, non-workspace post.
author_id	uuid	- FK → "Users".users(users_uuid) - ON DELETE CASCADE	NOT NULL	User who created the post.
title	text	—	nullable	Optional internal title for the post.
content	text	CHECK content_required_on_submit	NOT NULL, default ''	The post body text. Can be empty only while status is 'draft'; once submitted beyond draft, trimmed content must be non-empty.
status	post_statuses_enum	—	NOT NULL, default 'draft'	Editorial workflow stage: draft, pending_review, rejected, ready_for_distribution, published, or failed.
prompt_template_id	uuid	- FK→workspaces.prompt_templates (id) - ON DELETE SET NULL	nullable	Prompt template used to AI-generate this post, if any.
knowledge_base_id	uuid	- FK→workspaces.knowledge_base_documents(id) - ON DELETE SET NULL	nullable	Knowledge-base document used as context for generation, if any.

ai_generated	boolean	—	NOT NULL, default false	Whether the content was produced by AI generation rather than written manually.
seo_keywords	jsonb	—	nullable	SEO keyword data stored as JSON
seo_hashtags	jsonb	—	nullable	SEO hashtag data stored as JSON.
submitted_at	timestamp t	—	nullable	Timestamp the post was submitted for review.
reviewed_by	uuid	FK → "Users".users(users_uuid)	nullable	Reviewer who approved/rejected the post, if reviewed.
reviewed_at	timestamp t	—	nullable	Timestamp of the review decision.
reject_reason	text	—	nullable	Explanation given when the post is rejected.
published_at	timestamp t	—	nullable	Timestamp the post was published
created_at	timestamp t	—	NOT NULL, default now()	Creation timestamp.
updated_at	timestamp t	—	NOT NULL, default now()	Last-update timestamp; auto-maintained by trg_posts_updated_at.

4.2.4.7 post_platforms

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique identifier for the post-to-platform publishing record.
post_id	uuid	- FK → workspaces.posts(id) - ON DELETE CASCADE - UNIQUE with social_account_id	NOT NULL	The post being published. Combined with social_account_id, must be unique (a post cannot target the same account twice).
social_account_id	uuid	- FK→workspaces.social_accounts(social_acc_id) - ON DELETE RESTRICT	NOT NULL	The connected account being published to; cannot be deleted while publishing records reference it.
platform	platform_enum	—	NOT NULL	Denormalized copy of the target platform, for quick filtering without a join.
publish_status	publish_status_enum	—	NOT NULL, default 'pending'	Outcome of the publish attempt for this specific account: pending, published, or failed.
platform_post_id	text	—	nullable	ID of the resulting post as returned by the external platform's API, once published.
platform_post_url	text	—	nullable	Public URL of the published post on the external platform.
retry_count	smallint	—	NOT NULL, default 0	Number of times publishing has been retried after failure.

error_message	text	—	nullable	Error detail captured if the publish attempt failed.
published_at	timestamp	—	nullable	Timestamp this specific platform publish succeeded.
created_at	timestamp	—	NOT NULL, default now()	Timestamp this publishing record was created.

4.2.4.8 post_media

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique identifier for the media item.
post_id	uuid	- FK→workspaces.posts(id) - ON DELETE CASCADE	NOT NULL	The post this image belongs to.
image_url	text	—	NOT NULL	URL of the image.
position	smallint	—	NOT NULL, default 0	Display order of the image within the post's media gallery (0 = first).
uploaded_at	timestamp	—	NOT NULL, default now()	Timestamp the image was uploaded.

4.2.4.9 post_reviews

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique identifier for the review record.

post_id	uuid	- FK → workspaces.posts(id) - ON DELETE CASCADE	NOT NULL	The post being reviewed.
reviewer_id	uuid	FK → "Users".users(users_uuid)	NOT NULL	The user who performed the review.
action	review_action_enum	—	NOT NULL	The decision made: approve or reject.
comment	text	—	nullable	Optional reviewer feedback.
created_at	timestamp t	—	NOT NULL, default now()	Timestamp the review decision was recorded.

4.2.4.10 post_analytics

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique identifier for the analytics snapshot.
post_platform_id	uuid	- FK→workspaces.post_platforms(id) - ON DELETE CASCADE	NOT NULL	The specific platform publishing record these metrics belong to.
likes	integer	—	NOT NULL, default 0	Like count at the time of fetch.
comments	integer	—	NOT NULL, default 0	Comment count at the time of fetch.
shares	integer	—	NOT NULL, default 0	Share count at the time of fetch.
impressions	integer	—	NOT NULL, default 0	view count at the time of fetch.
is_cached	boolean	—	NOT NULL, default false	Whether this snapshot is a cached value rather than a fresh API fetch.

fetches_at	timestamp t	—	NOT NULL, default now()	Timestamp this snapshot was fetched.
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4.2.4.11 tasks

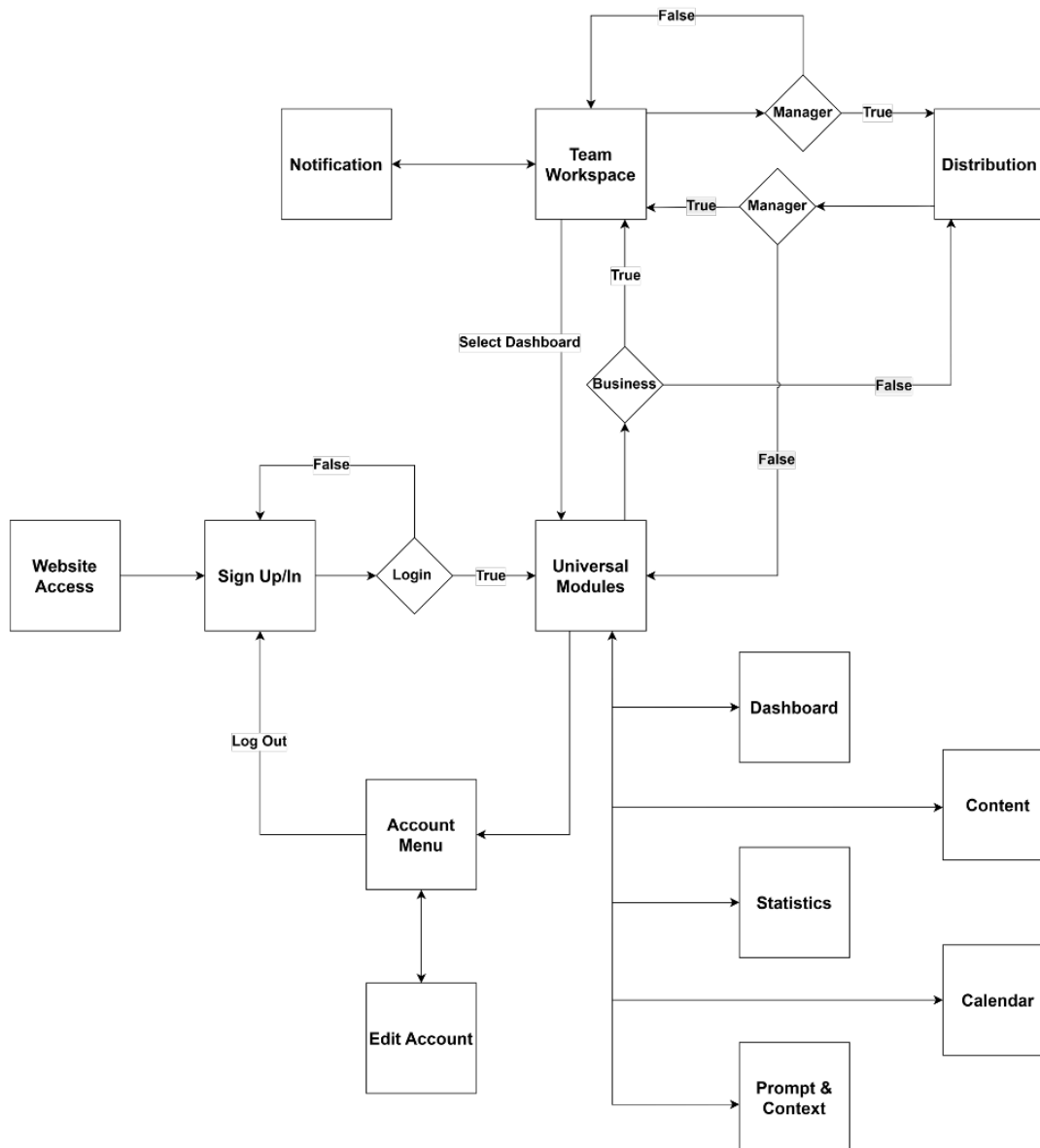
Attribute	Data Type	Key / Constraint	Nullable / Default	Description
id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique, time-ordered identifier for the task.
workspace_id	varchar(16)	- FK → workspaces.workspaces(workspace_uuid) - ON DELETE CASCADE	NOT NULL	Workspace the task belongs to. Deleting the workspace deletes all of its tasks.
title	text	—	NOT NULL	Short title describing the task.
content	text	—	NOT NULL, default ''	Longer free-text description/body of the task.
status	task_status_enum	—	NOT NULL, default 'todo'	Current workflow stage: todo, in_progress, review, completed, or cancelled.
priority	task_priority_enum	—	NOT NULL, default 'medium'	Urgency level: low, medium, high, or urgent.
assigned_to	uuid	- FK → "Users".users(users_uuid) - ON DELETE SET NULL	NOT NULL	The user responsible for the task.
created_at	timestamp tz	—	NOT NULL, default now()	Timestamp the task was created.
updated_at	timestamp tz	—	NOT NULL, default now()	Timestamp of the last update; auto-maintained by trg_tasks_updated_at.

4.2.4.12 task_attachments

Attribute	Data Type	Key / Constraint	Nullable / Default	Description
id	uuid	PRIMARY KEY	NOT NULL, default uuidv7()	Unique identifier for the attachment record.
task_id	uuid	• FK → workspaces.tasks(id) • ON DELETE CASCADE	NOT NULL	The task this attachment belongs to. Deleting the task deletes its attachments.
image_url	text	—	NOT NULL	URL path of the attached image.
uploaded_at	timestampz	—	NOT NULL, default now()	Timestamp the attachment was uploaded.

5 User Interface and User Experience Design

5.1 Screen Diagram



Seq	Screen	Description
1	Website Access	The entry point of the Omni Platforms system. A user reaches this screen by opening the web application in a browser that supports HTML5 over HTTPS; from here they are directed to the Sign Up/In screen to authenticate. Available to all users, before any role is assigned.
2	Sign Up/In	The combined registration and login screen. New users register with an email, password, username and Account Type (Individual or Business, the latter further split into Manager or Member), while returning users sign in with their credentials. A failed login returns the user to this screen. Available to all users, before any role is assigned.
3	Universal Modules	The shared post-login hub reached once a user authenticates successfully (Login = True). It provides the primary side navigation to the Dashboard, Content, Statistics, Calendar and Prompt & Context modules for all roles, and, depending on whether the account is a Business account and whether the user is a Manager, additionally gates access to the Team Workspace and Distribution modules.
4	Dashboard	The main landing/overview screen, showing a personalised summary of interactions, assigned

		tasks, calendar items and connected channels. Its widgets are role-dependent: Managers additionally see the Approval Requests queue and the team's Assigned Tasks ledger; Members see a My Submissions queue and a Tasks Done progress bar instead of team-wide data; Individuals see a simplified, team-free layout with a personal monthly goal.
5	Content	The Create & Edit Post screen, where users write posts manually or with AI assistance (using saved prompts and knowledge-base context), attach media, apply GEO/SEO optimisation, and select target platforms. A Member can only Save as Draft or Submit for Review and cannot publish directly; a Manager or Individual can send approved/self-authored content straight to distribution.
6	Statistics	The Statistical Report & Analytics screen. It displays engagement bar charts, platform-share breakdowns, and AI-generated performance reports built from Facebook/LinkedIn interaction data, along with a history of previously saved reports. Manager and Member views are scoped to the shared workspace; Individual views show only the user's own personal metrics.

7	Calendar	Displays scheduled tasks and deadlines through a monthly grid and an upcoming-deadline timeline. Managers additionally see a "Tasks Assigned to Others" summary card and the full four-colour (including team) priority legend; Members see the team-coded legend without the delegation summary; Individuals see only a three-colour personal priority legend, since delegation and team tasks do not apply to them.
8	Prompt & Context	A library screen for creating, storing and reusing AI prompt templates and knowledge-base context documents, which are consumed by the AI-assisted content generation feature on the Content screen. Available to all roles.
9	Team Workspace	The Business workspace-management hub. It is reached only when the account is a Business account and the Manager check succeeds or a Member is logged in; Individual accounts never reach this screen, since they have no team to manage. Managers use it to create and configure the workspace, review and approve/reject Members' draft posts, enable/disable distribution channels for the workspace, assign tasks, and manage the member roster. Members use it to view their

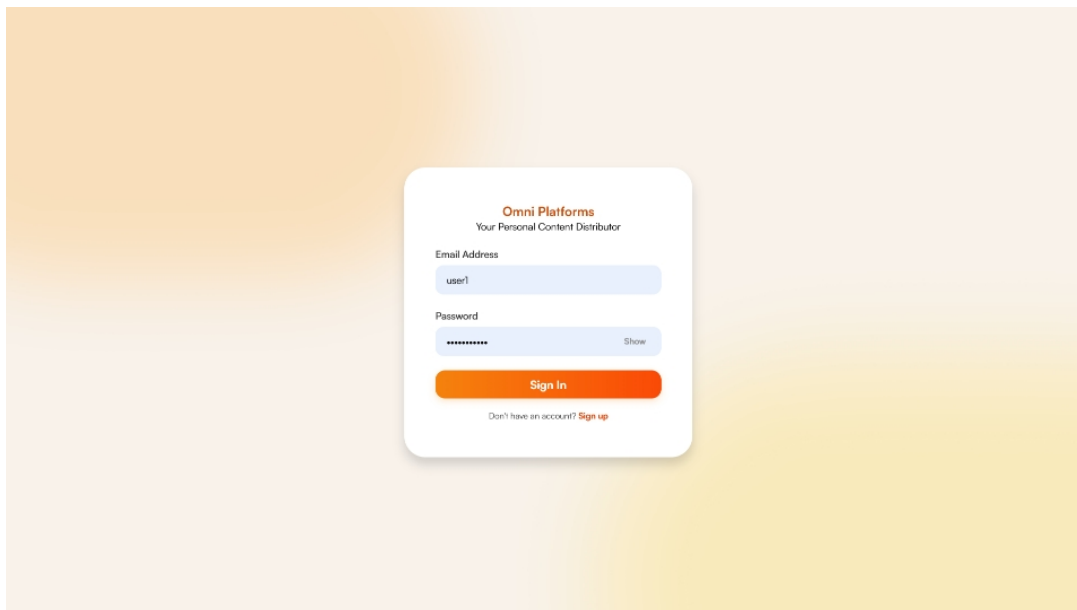
		own submission status and a read-only list of the channels the Manager has enabled for them.
10	Notification	Shows real-time alerts, such as new approval requests, task assignments and review outcomes (approved or rejected). This module is exclusive to Business accounts (Manager and Member); Individual users do not have a Notification screen, since they do not go through an approval workflow.
11	Account Menu	The menu reached from the user's profile/avatar icon, offering account-level actions such as opening Edit Account or Logging Out (which returns the user to the Sign Up/In screen). Available to all roles.
12	Edit Account	The profile-editing screen where a user updates personal information, namely Display Name, Avatar and Time Zone. Available to all roles.
13	Distribution	The channel-connection screen for linking, viewing, enabling/disabling and removing Facebook and LinkedIn accounts through OAuth authorisation. Managers and Individuals have full control, including adding and removing channels (and, for Managers,

		toggling a channel on or off for the whole workspace). Members cannot access this screen's configuration controls; they may only publish to the channels that a Manager has already enabled for their workspace.
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5.2 Screen Specifications

5.2.1 Screen “Sign In”

* Screen Designs



* Presentation Format

- Layout & Structure:
 - A modal card centered horizontally and vertically on a full-screen soft gradient background (cream to warm yellow/orange).
 - Rounded corners (border-radius) with a subtle drop-shadow to create elevation above the background.
- Header Elements:
 - Brand Title: Bold, orange typography displaying "Omni Platforms".

- Subtitle: Neutral dark grey text underneath reading *"Your Personal Content Distributor"*.
- Form Controls:
 - Email Address Input Field: Light blue/grey rounded text box with a top label (Email Address).
 - Password Input Field: Light blue/grey rounded input box with a top label (Password), rendering password characters as masked dots by default.
 - Password Visibility Toggle: Text button (Show / Hide) positioned inside the right end of the Password field.
- Action Controls:
 - Sign In Button: Full-width rounded call-to-action button with a vibrant orange background and white text reading "Sign In".
 - Registration Prompt: Footer link text at the bottom reading *"Don't have an account? "* with a highlighted, clickable "Sign up" link in orange.

* Event Handling

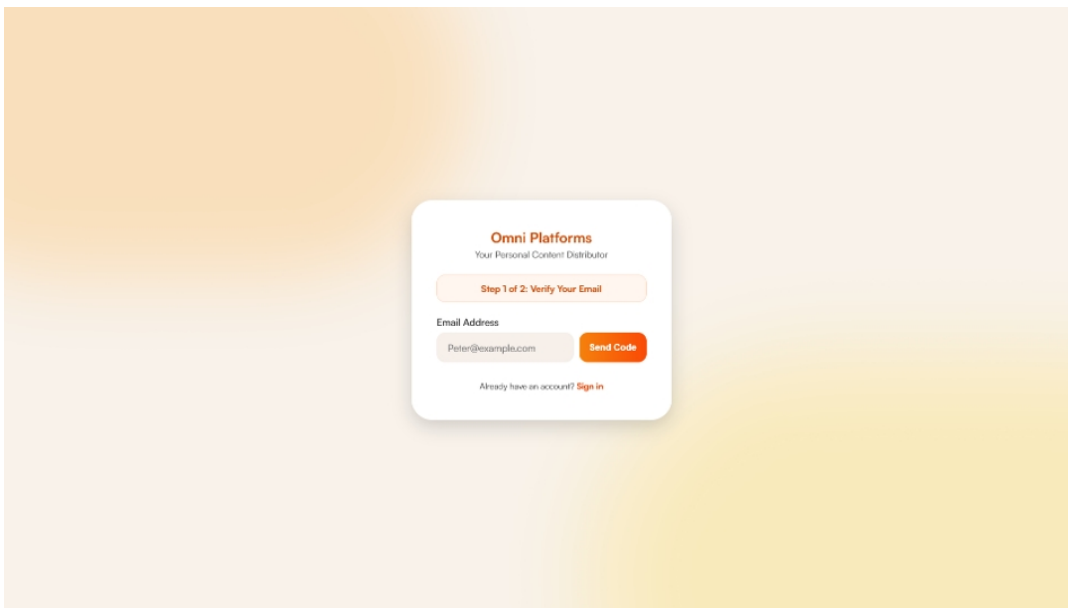
Control Element	Event Type	Action & Behavior
Email Address Input	onChange onInput	Captures user text input for the account login address or username. Validates standard format upon focus loss or form submission.
Password Input	onChange onInput	Captures keypresses for account authentication. Text is masked using default bullet characters (•).
"Show / Hide" Toggle	onClick	Toggles the input field attribute between type="password" and type="text". Updates button label dynamically between "Show" and "Hide".

"Sign In" Button	onClick onSubmit	/	1. Triggers form validation to check for non-empty and valid inputs. 2. Initiates the authentication request (POST /api/auth/login) with input credentials. 3. On success, stores the JWT session token and redirects the user to the Dashboard. 4. On failure, displays an inline or toast error message (e.g., <i>"Invalid email or password"</i>).
"Sign up" Link	onClick		Redirects the user from the Login screen to the Sign Up / Registration Screen.

5.2.2 Screen "Sign Up"

* Screen Designs

Step 1: Email Input (Send PIN)



Step 1.1: PIN Code Entry & Resend Timer

Omni Platforms
Your Personal Content Distributor

Step 1 of 2: Verify Your Email

Verification PIN sent to user1@example.com!
(Demo PIN: 123456)

Email Address
user1@example.com

Enter PIN Code
1 2 3 4 5 6

Verify & Continue

Didn't receive code? Resend in 57s

Already have an account? [Sign in](#)

Step 2a: Account Information Completion (Business Mode)

Omni Platforms
Your Personal Content Distributor

user1@example.com [Edit Email](#)

Personal **Business**

Username
How should we call you?

You are? **Manager** Member

Password
Enter your password Show

Workspace name
Workspace name

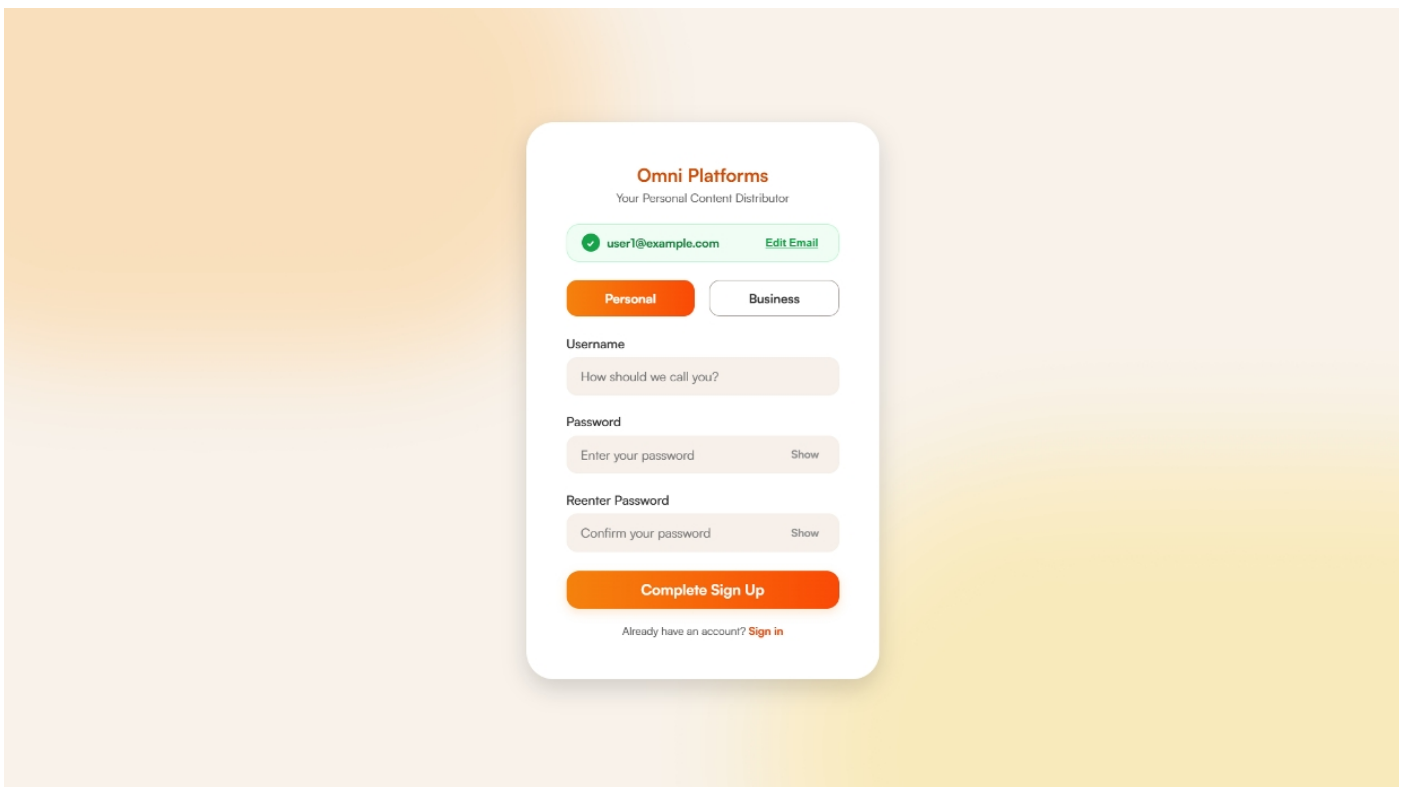
Reenter Password
Confirm your password Show

Workspace password
4-8 digit PIN Show

Complete Sign Up

Already have an account? [Sign in](#)

Step 2b: Account Information Completion (Personal Mode)



* Presentation Format

Layout & Alignment:

- A centered modal card on a soft warm gradient background (cream transitioning to pale yellow/orange).
- The modal width dynamically adjusts: a compact single-column layout during initial verification and Personal registration, expanding into a two-column grid for Business mode setup.

Header Branding & Step Progress Indicator:

- Brand Header: Bold orange text reading "Omni Platforms" with subtitle *"Your Personal Content Distributor"*.
- Step Banner: A soft orange pill container displaying step progression (e.g., Step 1 of 2: Verify Your Email).

Verified Email Status Badge (Step 2):

- Light green background banner displaying a green checkmark icon, the confirmed email address (user1@example.com), and an "Edit Email" link.

Input Fields & Controls:

- Email Field & Send Action: Input box with an inline "Send Code" orange button.

- PIN Code Field: Center-aligned input field formatted for multi-digit PIN entry (1 2 3 4 5 6).
- Account Type Switcher: Segmented toggle for "Personal" and "Business" account modes.
- Business Role Selector: Segmented toggle for "Manager" and "Member" roles.
- Standard Text Fields: Rounded inputs for Username (How should we call you?), Password, Reenter Password, Workspace Name, and Workspace Password (4-8 digit PIN).

Action Buttons & Links:

- Verify & Continue / Complete Sign Up Buttons: Full-width orange primary action buttons.
- Resend PIN Link: Secondary action link with a dynamic 60-second countdown timer (Resend in 57s).
- Sign In Footer: Footer navigation link reading *"Already have an account? Sign in"*.

* Event Handling

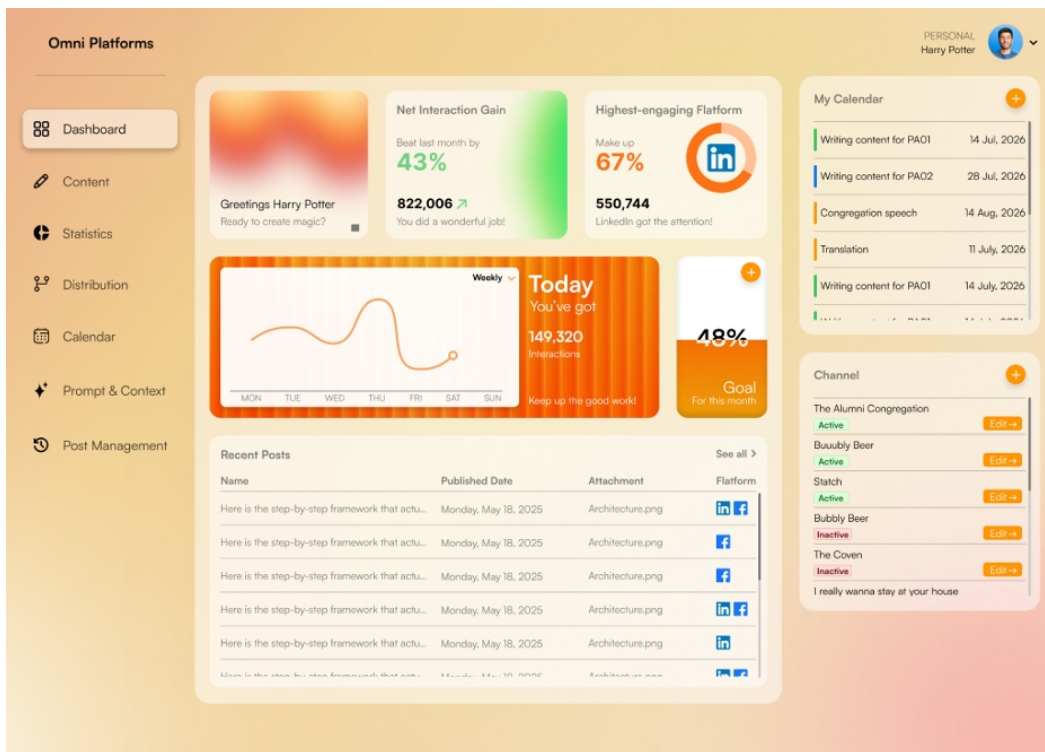
Control / Element	Event Type	Action & Behavior
Email Address Input	onChange / onInput	Captures user email address. Validates RFC standard format on blur or click.
"Send Code" Button	onClick	Sends a request to deliver a 6-digit verification PIN to the specified email. Reveals PIN input controls and starts the resend timer
Enter PIN Code Input	onChange / onInput	Accepts numeric keypresses (up to 6 digits).
"Verify & Continue" Button	onClick / onSubmit	Submits the entered PIN. On successful verification, transitions to Step 2 (Account Setup).

"Resend Code" Link	onClick	Re-triggers the verification email and resets the countdown timer to 60 seconds. Disabled while timer is active.
"Edit Email" Link	onClick	Returns the user back to the very first step - the "Email Address Input" step - to change the email address and clear the verified badge.
"Personal" / "Business" Toggle	onClick	• Personal: Swaps layout to a single column, hiding workspace parameters. • Business: Expands layout to two columns, revealing role selection, Workspace Name, and Workspace Password fields.
"Manager" / "Member" Toggle	onClick	• Manager: Configures workspace fields for creating a new workspace. • Member: Configures workspace fields for joining an existing workspace.
"Show / Hide" Toggles	onClick	Toggles input visibility between masked dots and plain text for password fields.
"Complete Sign Up" Button	onClick / onSubmit	1. Executes frontend validation (checks matching passwords and required fields). 2. Transmits payload (POST /api/auth/register) 3. On Success: Stores session token and redirects to the Dashboard.

5.2.3 Screen "Dashboard"

* Screen Designs

Individual view:



Manager & Member view (Business):



* Presentation Format

- Global Layout & Structure:
 - Sidebar Navigation (Left): Persistent vertical navigation menu with links to app modules. Includes dynamic visibility based on role (e.g., Team Workspace item is hidden in Personal mode).
 - Top Utility Bar (Right): Displays role indicator badge (MANAGER, MEMBER, or PERSONAL), user avatar, profile drop-down menu, and notification bell icon. The Member view adds a top header displaying the active Workspace ID and manager attribution.
- Central Content Grid:
 - Top Metric Cards Row:
 - Greetings Banner: Personalized welcome card ("*Greetings [UserName]*").
 - Net Interaction Gain: High-level KPI metric with growth rate (+43%) and total count (822,006).
 - Highest-Engaging Platform: Channel distribution widget highlighting top platform share (e.g., LinkedIn 67%).
 - Analytics & Progress Row:
 - Interaction Timeline Chart: Interactive line chart with time range selector dropdown (Weekly / Monthly) paired with real-time interaction totals.
 - Goal Tracker Card: Visual percentage indicator (e.g., 48% Goal achieved).
 - Lower Ledger Area:
 - Tasks Ledger (Manager / Member): Displays Assigned Tasks (Manager) or My Tasks (Member) with priority tags (Low, Medium, High), dates, attachments, and assignees/owners.
 - Recent Posts Matrix: Ledger tracking recently published content across target platforms.
- Right Utility Sidebar:
 - Role-Specific Action Panel (Top Widget):
 - Manager View: Approval Requests list with clickable "View request" links.

- Member View: My Submissions queue with status badges (Pending, Denied, Approved).
- Personal View: Omitted; Calendar widget moves to top priority.
- My Calendar Widget: Color-coded upcoming schedule entries with a quick-add "+" action icon.
- Channel Management Widget: Connected social profiles list with status tags (Active, Inactive) and direct "Edit →" or action controls.

* Event Handling

Control / Element	Event Type	Action & Behavior
Sidebar Menu Items	onClick	Navigates the application routing to the corresponding module screen (Dashboard, Content, Statistics, Distribution, Calendar, Team Workspace, Prompt & Context, Post Management).
Notification Bell Icon	onClick	Opens a popover listing recent notification alerts (e.g., post status updates, approval requests, task assignments).
User Profile Dropdown	onClick	Expands account option menu including Profile Settings, Workspace Details, and Logout trigger.
Timeline Timeframe Selector	onChange	Toggles chart data view between Weekly, Monthly, or Custom Range, triggering an API re-fetch for interaction metrics.
"View request" Links (Manager)	onClick	Opens the review modal/screen for the selected post draft, allowing the Manager to approve, reject, or leave comments.
"See all >" Buttons	onClick	Redirects to the complete dedicated management sub-screen (e.g., Post Management, Task Calendar, or Statistics Report Archive).

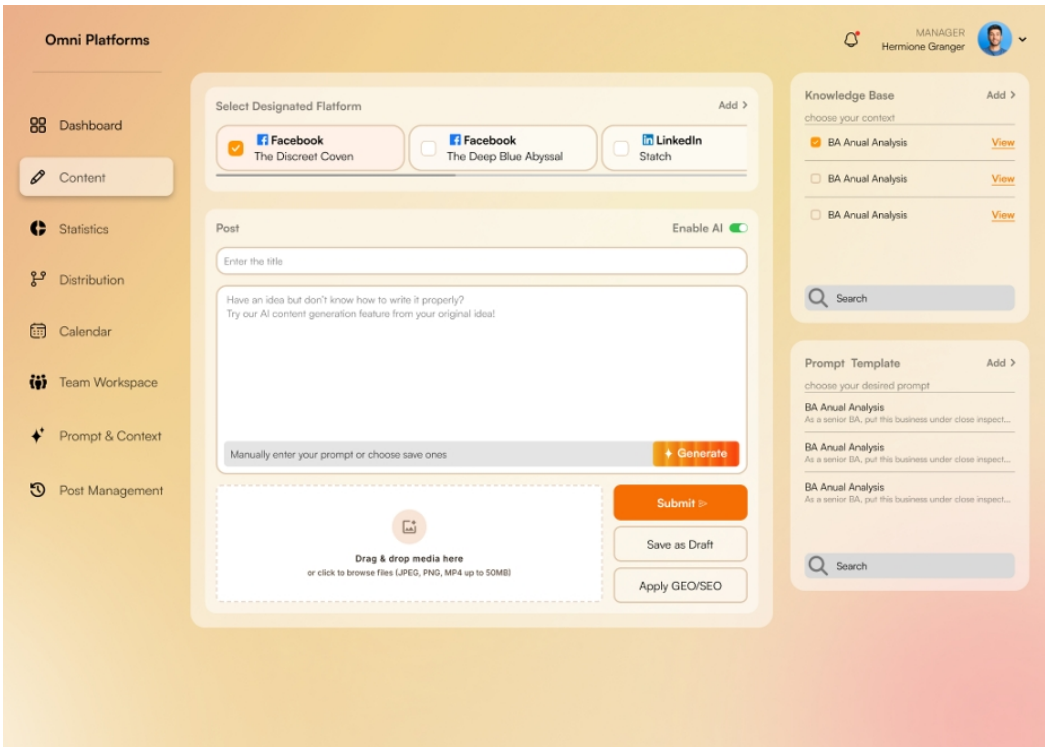
Calendar Quick Add Icon ("+")	onClick	Opens a modal overlay to quickly draft and schedule a new task or post event directly to the user's calendar.
Channel "Edit →" / Status Control	onClick	Opens account connection options or redirects to the Distribution management screen for OAuth reconfiguration.

5.2.4 Screen “Content”

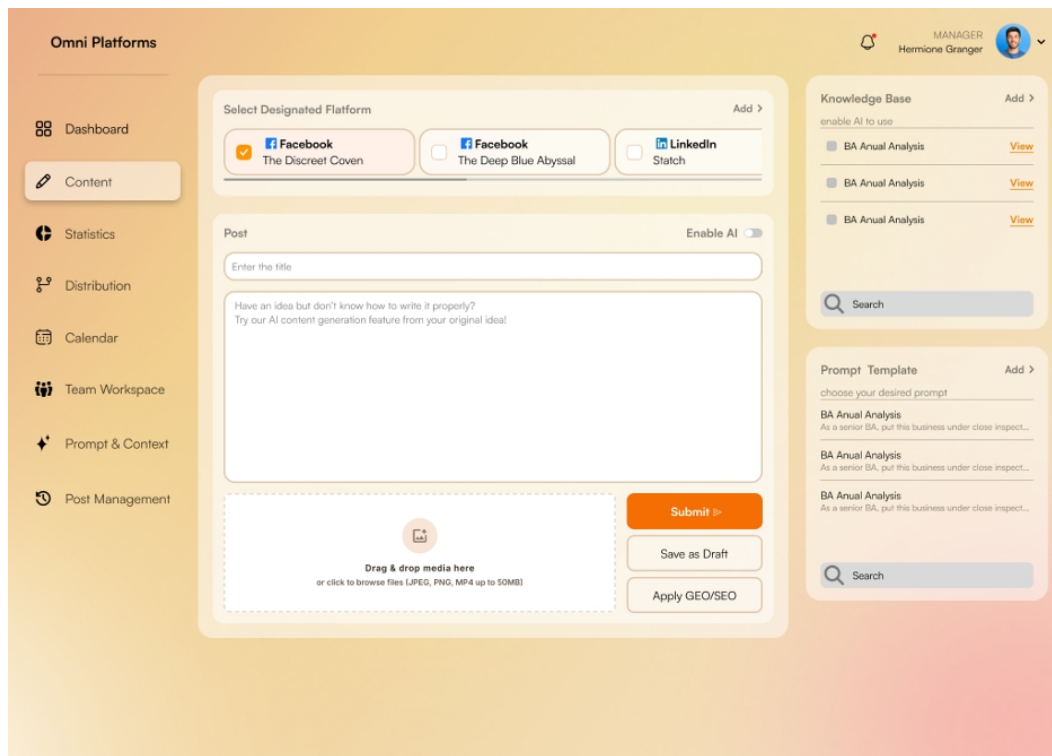
* Screen Designs

Universal view (generally the design stays consistent among 3 roles):

- AI-enabled view:



- AI-disenabled view:



* Presentation Format

- Layout & Structure:
 - Main Column (Left/Center): Contains target channel selection cards at the top, the post content editor with title/body fields in the middle, media drag-and-drop file upload zone at the bottom-left, and action submission buttons at the bottom-right.
 - AI Resource Sidebar (Right Column): Two stacked auxiliary panels containing selectable Knowledge Base contexts (top) and reusable Prompt Templates (bottom) equipped with dedicated search inputs.
- Select Designated Platform (Top Carousel Bar):
 - Horizontally scrollable target platform cards featuring checkboxes, social media network icons (e.g., Facebook, LinkedIn), page names, and an "Add >" quick link to attach new channels.
- Post Composition Workspace:
 - Header Controls: Section title (Post) paired with an Enable AI toggle switch.
 - Post Title Field: Single-line rounded input box (Enter the title).
 - Post Content Editor: Multi-line rich text area with default placeholder guidance (*"Have an idea but don't know how to write it properly? Try our AI content generation feature..."*).

- AI Prompt Action Bar (AI Enabled State): An embedded bottom text input inside the editor (Manually enter your prompt or choose saved ones) linked to a gradient "◆ Generate" button.
- Media Upload & Action Section:
 - Drag & Drop Media Area: Dashed drag-and-drop container accepting images and videos (JPEG, PNG, MP4 up to 50MB).
 - Primary Action Buttons:
 - "Submit »" Button: Vibrant orange primary action button.
 - "Save as Draft" Button: Secondary outline button.
 - "Apply GEO/SEO" Button: Optimization engine button.

* Event Handling

Control Element /	Event Type	Action & Behavior
Platform Selection Checkboxes	onClick	Toggles social media channel targets for the current post draft. Selected cards highlight with an active border and orange checkbox state.
"Add >" Channel Link	onClick	Redirects the user to the Distribution screen to connect or enable additional social media channels.
"Enable AI" Toggle Switch	onClick	• Toggle ON: Expands the embedded AI Prompt input bar with the "◆ Generate" button at the base of the editor and activates context options in the right sidebar. • Toggle OFF: Hides the AI prompt input bar, expanding total text editor space, and dims/disables right-sidebar AI context selections.
Post Input Title	onChange / onInput	Captures title text string for post indexing and management.

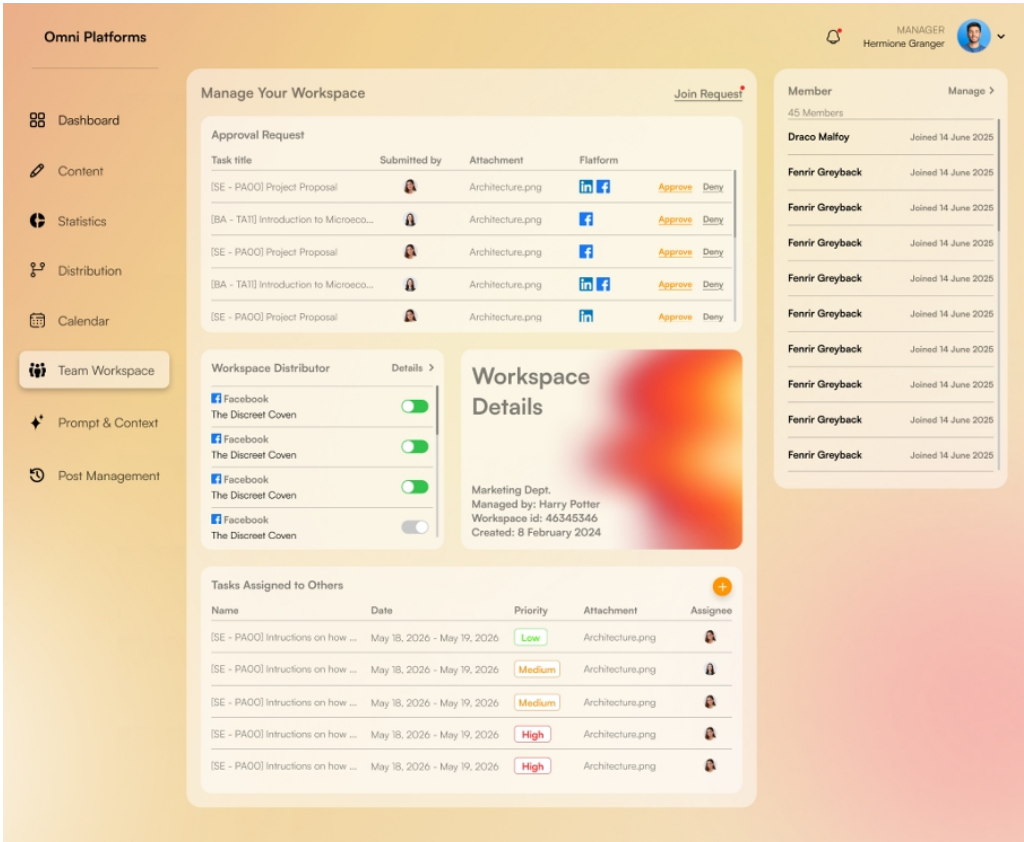
Post Content Textarea	onChange / onInput	Captures manual text composition, caption, or post body markup.
AI Prompt Input & "Generate" Button	onClick / onKeyDown	Reads selected Knowledge Base documents and Prompt Templates alongside the entered prompt, sending a request to the AI service (POST /api/ai/generate). Dynamically populates the generated text into the Post Content editor.
Drag & Drop Media Container	onDrop / onClick	Triggers native file selector or processes dropped files. Validates file format (JPEG, PNG, MP4) and checks size limits (≤ 50 MB) before displaying thumbnail previews.
"Submit" Button	onClick	• Manager / Individual Role: Validates inputs and sets post status to Ready for Distribution or publishes directly. • Member Role: Submits post draft into the Manager's review queue with status Pending Review.
"Save as Draft" Button	onClick	Saves current inputs, attached media, and assigned channels to Post Management under status Draft without triggering approval workflows.
"Apply GEO/SEO" Button	onClick	Analyzes post body text against GEO/SEO optimization algorithms, suggesting relevant search tags, hashtags, and structural enhancements.
Knowledge Base Checkboxes / "View"	onClick	• Checkbox: Attaches/detaches background context files to the active AI prompt. • View: Opens a modal preview displaying full grounding text/documents.

Prompt Template Selectors	onClick	Injects pre-configured prompt structures into the embedded AI prompt input bar.
Sidebar Search Bars	onChange / onInput	Filters Knowledge Base items or Prompt Templates by title string in real time.

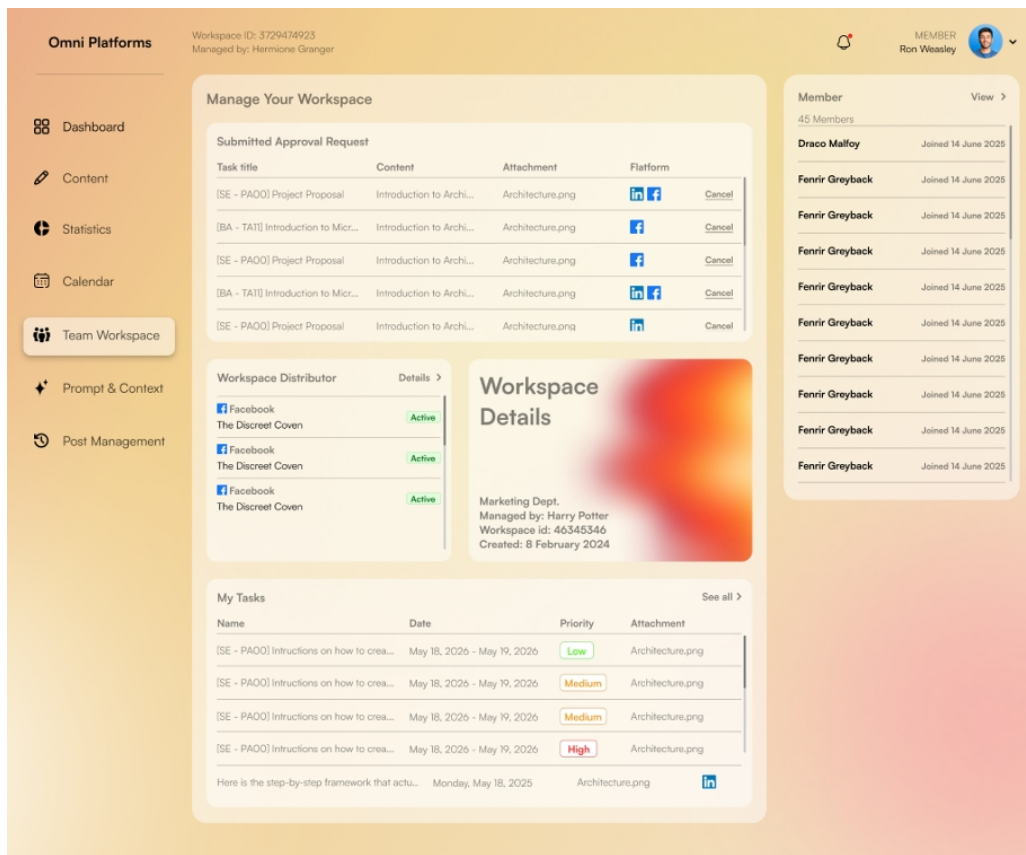
5.2.5 Screen “Team Workspace”

* Screen Designs

Manager view:



Member view:



* Presentation Format

- Main Work Area (Left/Center Column):
 - Header & Quick Actions: Screen title (Manage Your Workspace) featuring a "Join Request" alert trigger with a red notification dot at the top right (Manager view only).
 - Upper Moderation Ledger:
 - Manager View (Approval Request): Ledger listing posts submitted by team members, displaying Task Title, Author Avatar (Submitted by), File Attachments, Target Platforms, and direct action triggers (Approve / Deny).
 - Member View (Submitted Approval Request): Personal submission ledger displaying Task Title, Content Snippet, File Attachments, Target Platforms, and a "Cancel" request trigger.
 - Middle Information Cards:
 - Workspace Distributor Widget: Connected social media channels with enable/disable toggle switches (Manager view) or status tags

like Active (Member view). Includes a "Details >" link to jump to the Distribution module.

- **Workspace Details Card:** Visual gradient card displaying administrative workspace metadata, including Department, Workspace Manager, Workspace ID (46345346), and Creation Date.
- **Lower Task Ledger:**
 - **Manager View (Tasks Assigned to Others):** Task tracking table showing Task Name, Assigned Date Range, Priority Badges (Low, Medium, High), File Attachments, Assignee Avatar, and an orange "+" quick-create task button.
 - **Member View (My Tasks):** Personal assigned tasks matrix with Priority Badges, Dates, Attachments, and target platform icons.
- **Right Sidebar Panel:**
 - **Member List Widget:** Scrollable list displaying active workspace users, total member count (45 Members), join dates, and top action links ("Manage >" for Managers or "View >" for Members).

* Event Handling

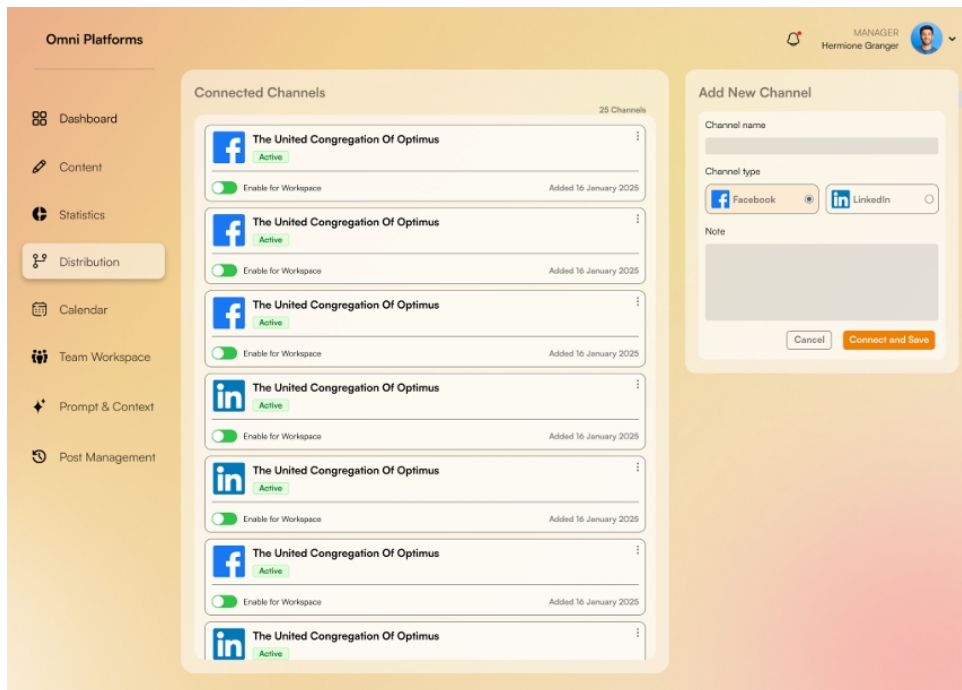
Control / Element	Event Type	Action & Behavior
"Join Request" Link (Manager View)	onClick	Opens a modal displaying incoming team member access requests with options to approve or reject new workspace joins.
"Approve" Button (Manager View)	onClick	Approves the Member's post submission, updating its status to Ready for Distribution or publishing it to target social channels immediately.
"Deny" Button (Manager View)	onClick	Rejects the Member's post draft and opens a review comment modal to supply feedback/rejection reasons back to the author.
"Cancel" Button (Member View)	onClick	Revokes a previously submitted post review request, restoring the post status back to editable Draft.

Workspace Distributor Toggle Switch (Manager)	onClick	Enables or disables target social channel availability across the workspace for team members.
Workspace Distributor "Details >"	onClick	Redirects to the Distribution screen to add, re-authenticate, or remove social accounts.
Task Quick Add Button ("+")	onClick	Opens a modal to create a new task, assign priority, set deadline dates, attach files, and delegate it to a specific team member.
Member Panel "Manage >" Link	onClick	Expands member administrative management views to update team roles or remove users from the workspace.

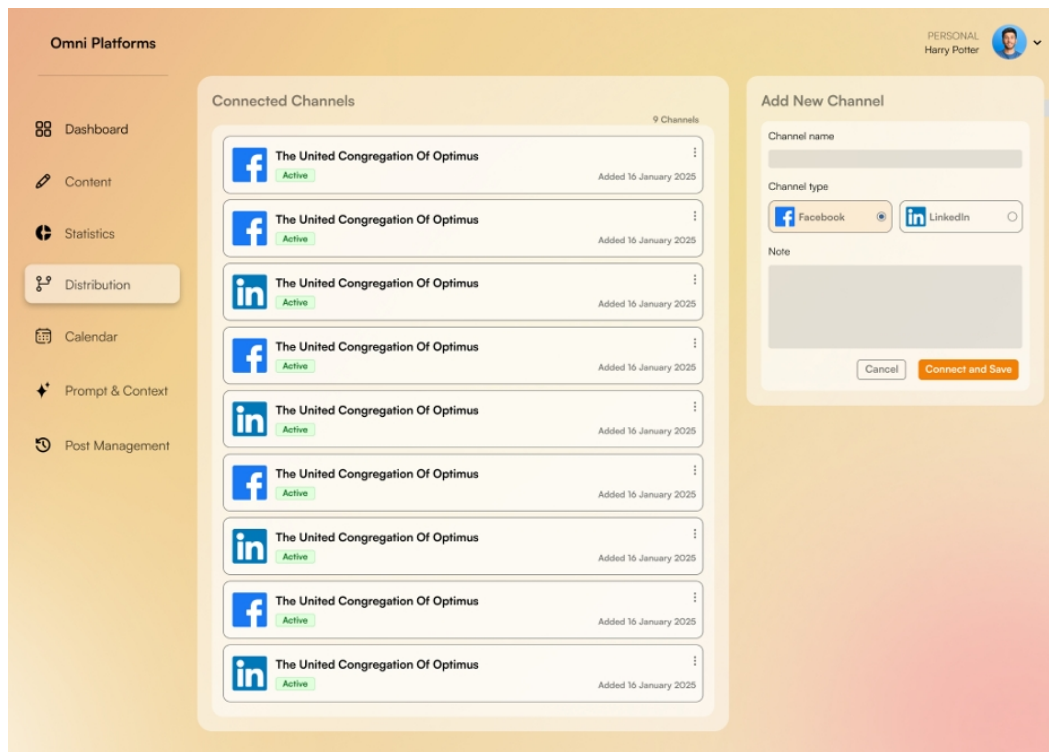
5.2.6 Screen "Distribution"

* Screen Designs

Manager view:



Individual view:



* Presentation Format

- Main Column (Connected Channels List - Left/Center):
 - Header: Section title (Connected Channels) featuring a total channel count badge (e.g., 25 Channels for Manager, 9 Channels for Personal).
 - Channel Card Layout: Vertical scrollable list of connected social media platforms (Facebook, LinkedIn).
 - Card Contents:
 - Platform Icon & Page Name: Network badge icon alongside the account/page handle.
 - Status Badge: Green Active or red Inactive pill indicator.
 - Context Options Menu: Three-dot vertical overflow icon (⋮) positioned at the top-right of each card.
 - Added Date Stamp: Displayed at the bottom-right (e.g., Added 16 January 2025).
 - Enable for Workspace Toggle (Manager View Only): A toggle switch labeled "Enable for Workspace" at the bottom-left of each channel card. Omitted in Personal view.
- Right Panel (Add New Channel Form):

- Form Header: Title reading Add New Channel.
- Channel Name Field: Single-line text input box (Channel name).
- Channel Type Selection: Radio button pair featuring platform choices with icons (Facebook and LinkedIn).
- Note Field: Multi-line text area for optional internal channel notes or descriptions.
- Action Controls: Secondary outline button (Cancel) and primary orange action button (Connect and Save).

* Event Handling

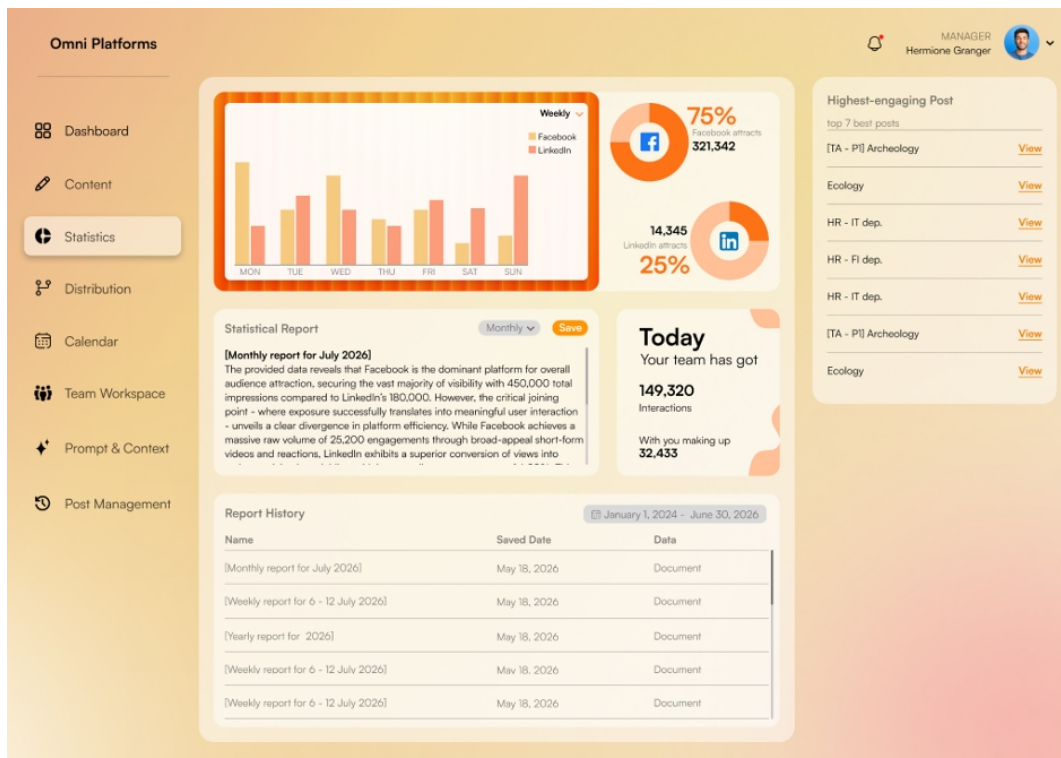
Control / Element	Event Type	Action & Behavior
"Enable for Workspace" Toggle (Manager View)	onClick	Grants or revokes access for team members in the workspace to distribute content through the specific social channel. Updates backend workspace channel permissions in real time.
Three-dot Menu (⋮)	onClick	Opens a popover menu with options to Re-authenticate OAuth, Edit Channel Details, or Disconnect / Delete Channel.
"Channel name" Input	onChange / onInput	Captures display label/nickname for the target channel being connected.
"Channel type" Radio Buttons	onClick	Selects the platform API protocol (Facebook OAuth 2.0 or LinkedIn OAuth 2.0).
"Note" Textarea	onChange / onInput	Captures optional contextual notes or internal operational details for the channel.

"Connect and Save" Button	onClick / onSubmit	1. Validates form inputs. 2. Initiates OAuth 2.0 authentication workflow via a popup or redirect to authorize platform API permissions. 3. On Auth Success: Stores authorization tokens, appends the new channel card to the Connected Channels list, and resets the form. 4. On Auth Failure: Displays an alert error toast (e.g., <i>"OAuth authorization failed"</i>).
"Cancel" Button	onClick	Resets all input fields in the Add New Channel form to their default empty states.

5.2.7 Screen "Statistics"

* Screen Designs

Universal view (generally the design stays consistent among 3 roles):



* Presentation Format

- Main Analytics & Reporting Area (Left/Center Column):

- Top Metrics Row:
 - Platform Engagement Bar Chart: Dual-bar visualization comparing platform interaction volumes (Facebook vs. LinkedIn) across days of the week, complete with a time range selector dropdown (Weekly / Monthly / Yearly).
 - Platform Breakdown Donut Rings: Proportion rings displaying share percentages (e.g., Facebook 75% with 321,342 interactions; LinkedIn 25% with 14,345 interactions).
- Middle Row (AI Statistical Report & Today Summary):
 - Statistical Report Card: Generative AI-synthesized narrative report evaluating platform performance and audience conversion trends. Features a report type dropdown (Monthly, Weekly, Yearly) and an orange "Save" button to archive the report.
 - "Today" Snapshot Card (Role-Dynamic Copy):
 - Manager View: Header reads *"Today - Your team has got"* (149,320 Interactions), followed by an individual baseline footer *"With you making up 32,433"*.
 - Member View: Header dynamically adjusts to focus purely on isolated personal contributions (*"Today - You've got"*), hiding overall team administrative baselines.
 - Personal View: Copy adjusts to *"Today - You've got"*. Total interaction counter reflects individual workspace metrics without team comparison footers.
- Lower Ledger (Report History Archive):
 - Ledger displaying previously saved AI reports, including Report Name, Saved Date, and Document Data download links. Equipped with a custom date-range picker filter (e.g., January 1, 2024 - June 30, 2026).
- Right Panel (Highest-Engaging Post):
 - Top 7 performing posts widget displaying title handles alongside direct "View" link triggers to inspect post performance details.

* Event Handling

Control / Element	Event Type	Action & Behavior

Bar Chart Timeframe Dropdown	onChange	Toggles visualization between Weekly, Monthly, and Yearly metric aggregation, triggering a backend data refetch for chart bars and donut percentage distributions.
Statistical Report Dropdown	onChange	Selects timeframe scope (Monthly, Weekly, Yearly) for the AI report generation engine.
Report "Save" Button	onClick	Archives the active AI-generated text summary into the Report History table, generating a new entry record with timestamp.
Report History Date Range Picker	onChange / onDateSelect	Filters the Report History table rows based on selected start and end dates.
"Document" Download Link	onClick	Exports or downloads the selected historical report as a PDF/text document file.
"View" Link (Highest-Enga ging Post)	onClick	Opens a detail popover/modal displaying granular interaction metrics (likes, shares, comments, reach) for that specific post draft.